Security Basics

Dr. PINAKI SANKAR CHATTERJEE

Introduction

... teaches us to rely not on the likelihood of the enemy's not coming, but on our own readiness to receive him; not on the chance of his not attacking, but rather on the fact that we have made our position unassailable.

—*The Art of War*, Sun Tzu

Outline

- Background
- Attacks, services and mechanisms
- Security attacks
- Security services
- Methods of Defense
- A model for Internetwork Security
- Internet standards and RFCs

Background

- Information Security requirements have changed in recent times
  - Traditionally provided by physical and administrative mechanisms
  - Many daily activities have been shifted from physical world to cyber space
    - Use of computers
      - Protect files and other stored information
    - Use of networks and communications links
      - Protect data during transmission
- The focus of many funding agencies in US
  - DOD, NSF, DHS, etc.
  - ONR: game theory for cyber security

Definitions

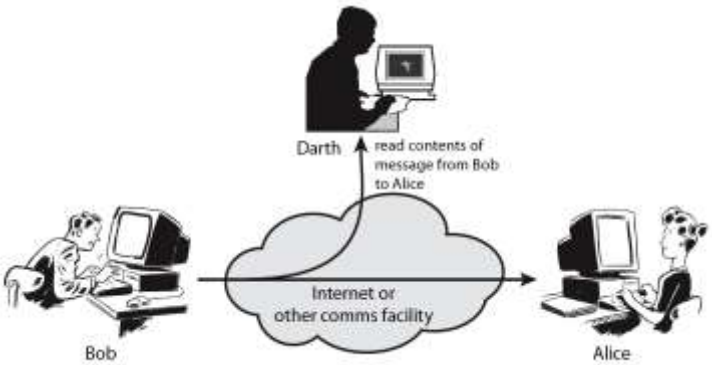
- Cybersecurity Security
  - Cybersecurity is the protection of information that is stored, transmitted, and processed in a networked system of computers, other digital devices, and network devices and transmission lines, including the Internet.
- Network Security
  - Measures to protect data during their transmission
- Information Security
  - This term refers to preservation of confidentiality, integrity, and availability of information.

3 Aspects of Info Security

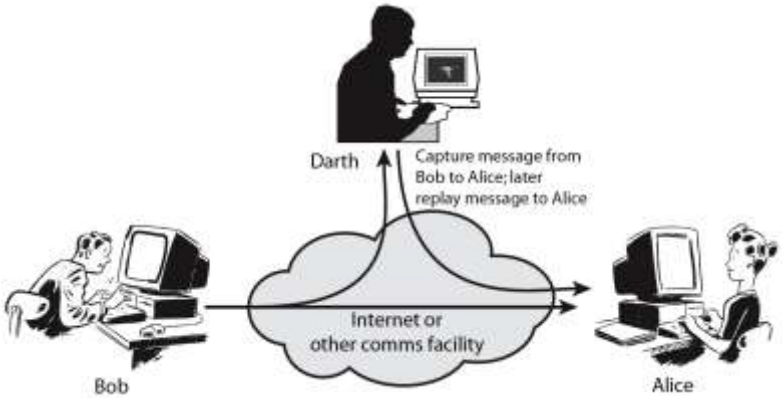
- **Security Attack**
  - Any action that compromises the security of information.
- **Security Mechanism**
  - A mechanism that is designed to detect, prevent, or recover from a security attack.
- **Security Service**
  - A service that enhances the security of data processing systems and information transfers.
    - Makes use of one or more security mechanisms.

Security Attacks

- Threat & attack
  - Often used equivalently
- There are a wide range of attacks
  - Two generic types
    - Passive



    - Active



Security Attack Classification

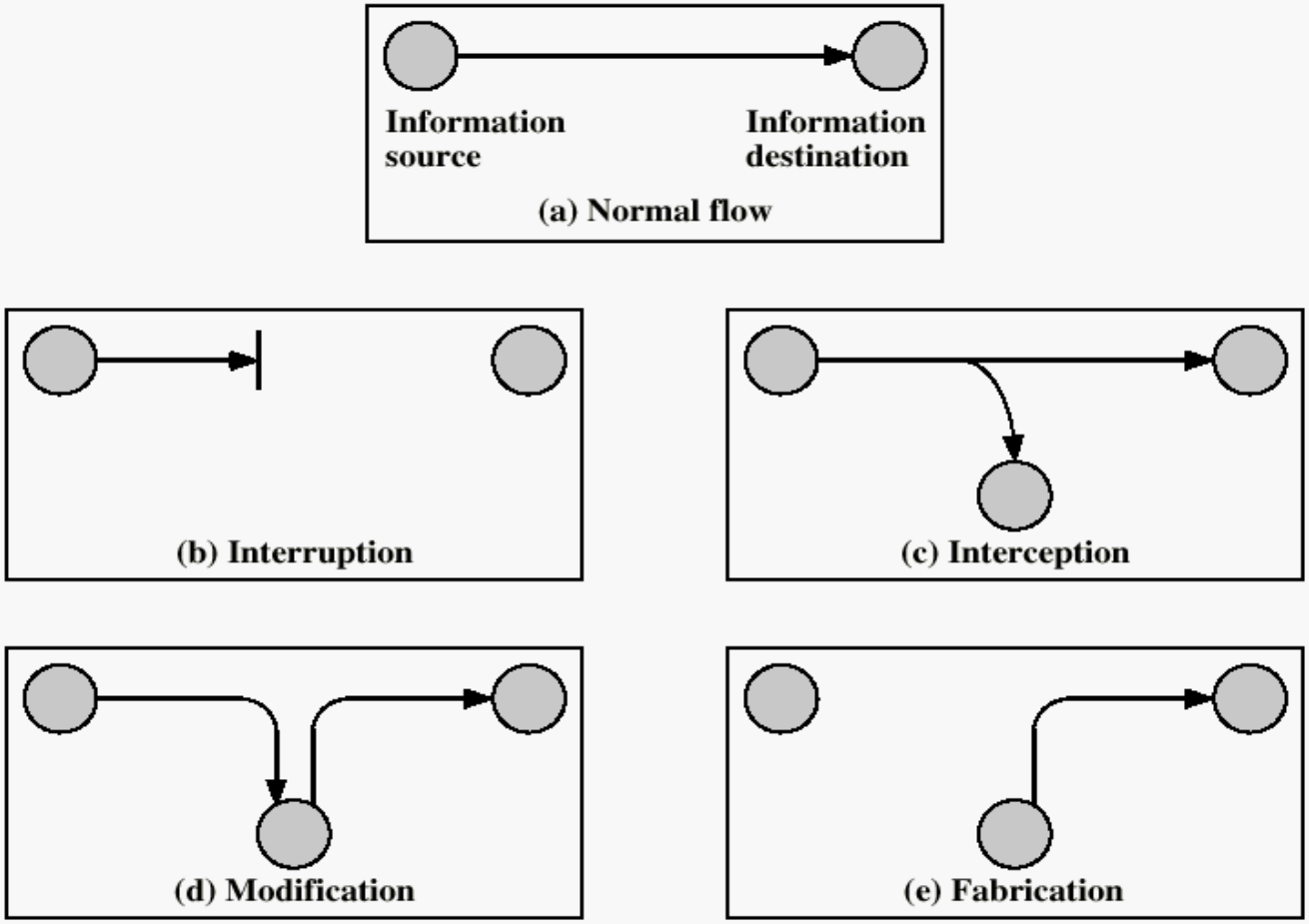
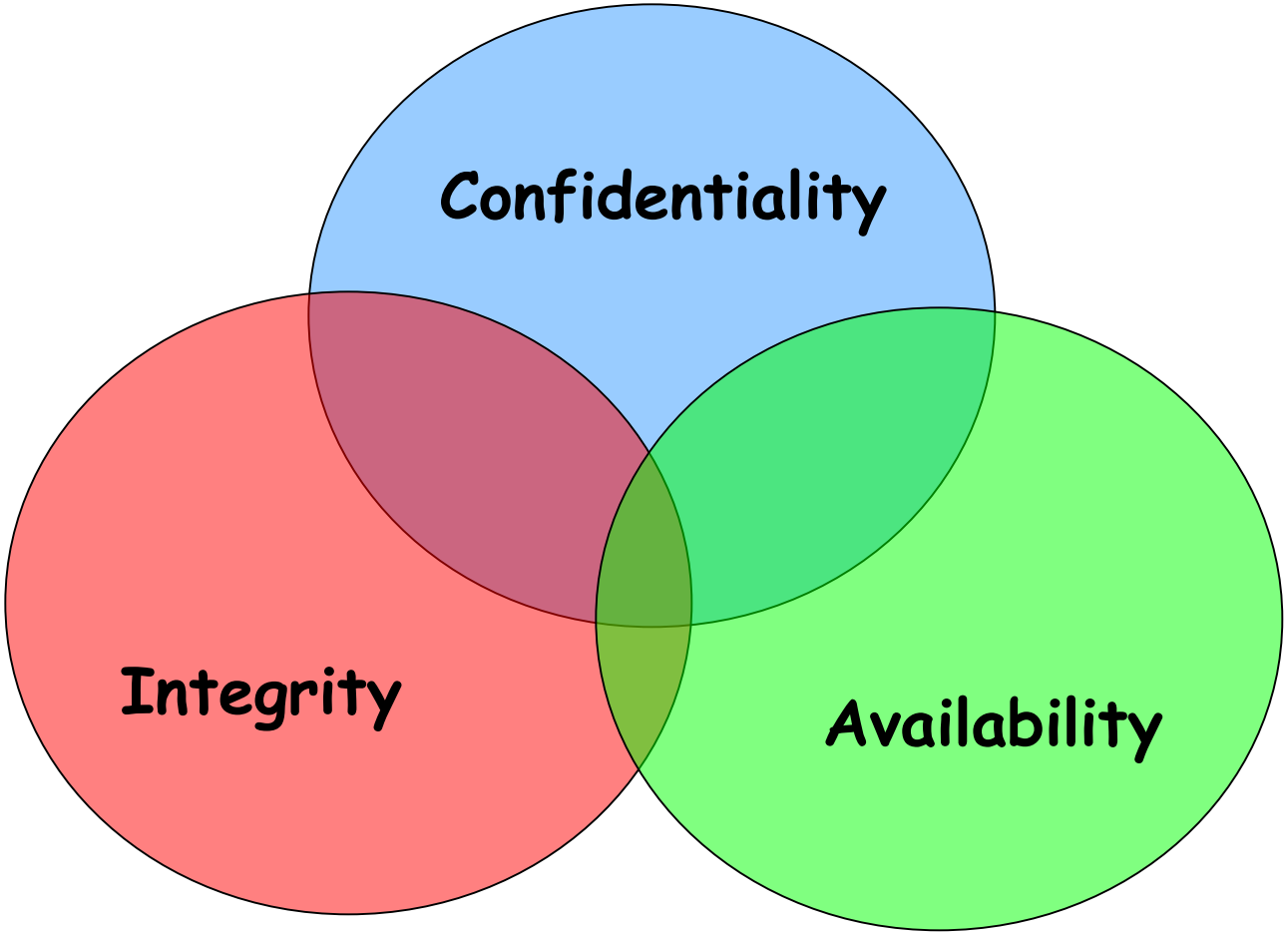


Figure 1.1 Security Threats

Security Attacks

- **Interruption:** This is an attack on availability
- **Interception:** This is an attack on confidentiality
- **Modification:** This is an attack on integrity
- **Fabrication:** This is an attack on authenticity

3 Primary Security Goals



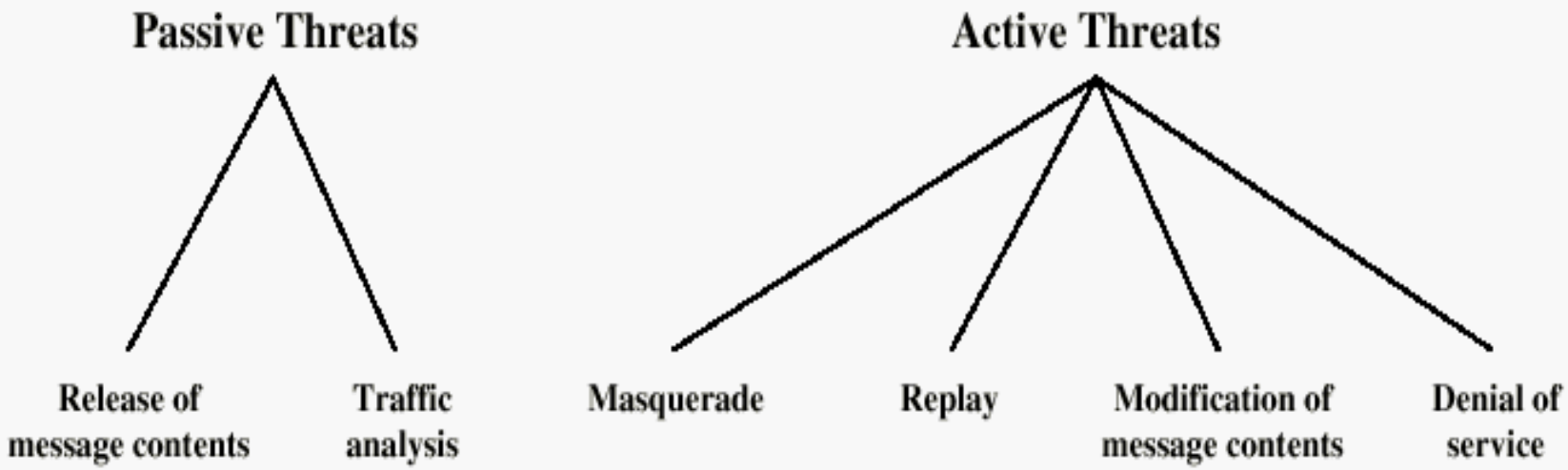


Figure 1.2 Active and Passive Security Threats

Security Services

X.800

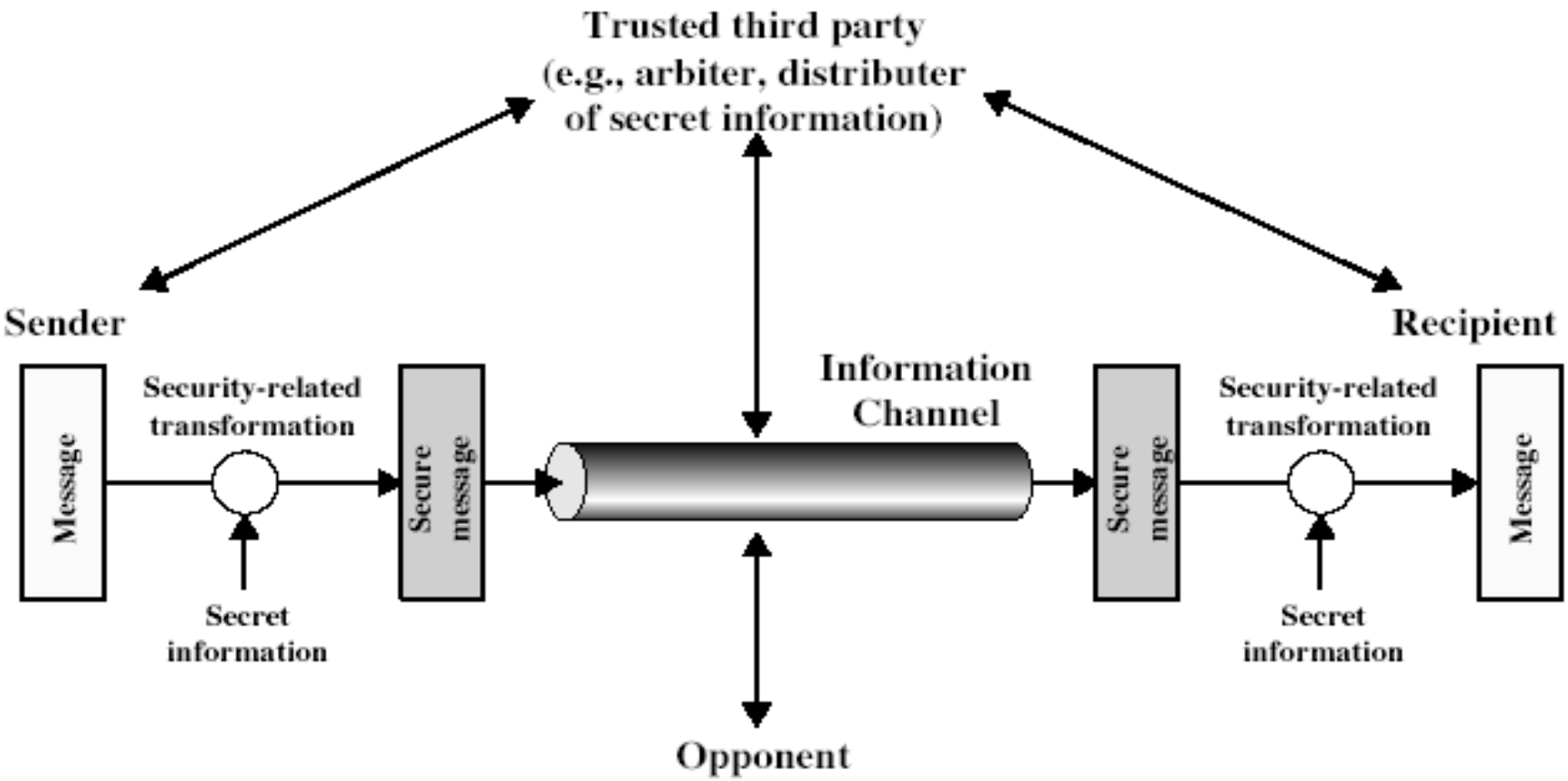
- A service provided by a protocol layer of communicating open systems, which ensures adequate security of the systems or of data transfers

- Confidentiality (privacy)
- Authentication (who created or sent the data)
- Integrity (has not been altered)
- Non-repudiation (the order is final)
- Access control (prevent misuse of resources)
- Availability (permanence, non-erasure)
  - Denial of Service Attacks
  - Virus that deletes files

Security Mechanism

- Features designed to detect, prevent, or recover from a security attack
- No single mechanism that will support all services required
- One particular element underlies many of the security mechanisms in use:
  - **Cryptographic techniques**
  - Hence we will focus on this topic first

Model for Network Security

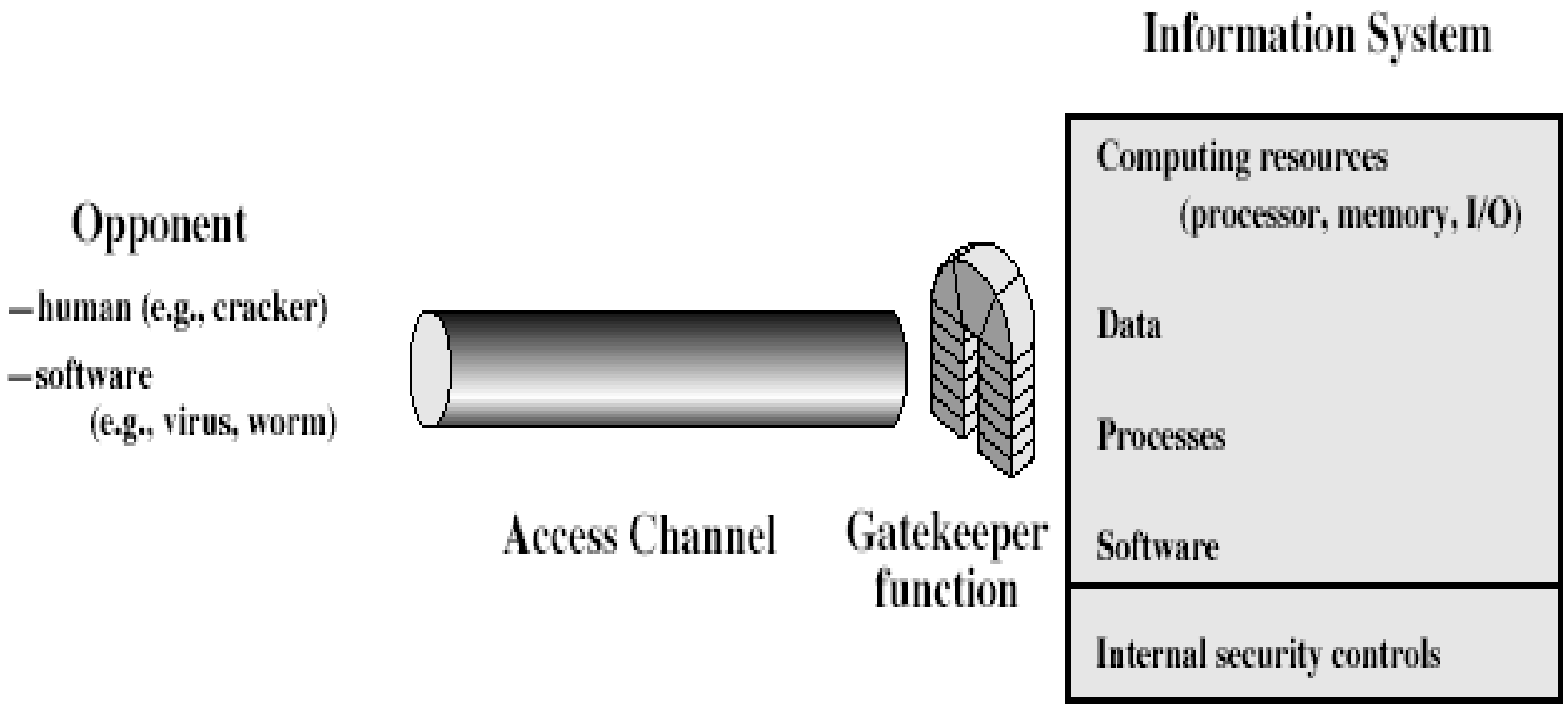


Model for Network Security

Using this model requires us to:

1. design a suitable algorithm for the security transformation (message de/encryption)
2. generate the secret information (keys) used by the algorithm
3. develop methods to distribute and share the secret information (keys)
4. specify a protocol enabling the principals to use the transformation and secret information for a security service (e.g. ssh)

Model for Network Access Security



Model for Network Access Security

Using this model requires us to implement:

1. Authentication
   - select appropriate gatekeeper functions to identify users
2. Authorization
   - implement security controls to ensure only authorized users access designated information or resources

Trusted computer systems may be useful to help implement this model

Methods of Defense

- Encryption
- Software Controls
  - Limit access in a database or in operating systems
  - Protect each user from other users
- Hardware Controls
  - Smartcard (ICC, used for digital signature and secure identification)
- Policies
  - Frequent changes of passwords
  - Recent study shows controversial arguments
- Physical Controls